

Research Article

Spatial assessment of tourists' demand for cultural ecosystem services in the Kalajun World Natural Heritage Site, China

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ABSTRACT

World Natural Heritage Sites encompass extraordinary natural phenomena and aesthetic values and play a prominent role in natural ecological protection and meeting human needs. However, the increased demand for ecotourism in the outer suburbs has caused World Natural Heritage Sites to face degradation risk or even loss of cultural ecosystem service (CES) function. Simultaneously, the academic community's spatial quantification of its CES needs and identification of its relationship with the natural environment remain unclear. Realistic requirements for protecting and developing World Natural Heritage Sites have brought many challenges to planning. Therefore, this study obtained point data related to tourist demand preferences through subjective participatory mapping and social media photos and combined them with the objective MaxEnt model to explore the spatial distribution patterns and contribution rate of tourists to the Kalajan CES demand and environmental variables at the grid scale. Furthermore, a Geodetector tool identified the key driving factors of tourists' CSE demands and the interactions between factors. The findings demonstrate that the aesthetic service experience is always the most favored and widely distributed among tourists. The tourists for high-demand CES areas are mainly concentrated in the south-central part of East Kalajun and along sightseeing vehicles. In addition, the spatial distribution patterns of aesthetics and recreational services are closely related to the vertical three-dimensional landscape of forest, grasslands, and other landforms embedded in the Kalajun tourist area. Elevation, accessibility, and Human footprint (HFP) significantly impact the spatial differentiation of tourist demand, and when combined with elevation and accessibility, the remaining driving factors are significantly enhanced.

Management implications

- 1) When improving traffic and accessibility, more attention should be paid to maintaining the original ecological environment and emphasizing natural experiences.
- 2) It is necessary to optimize the site selection of the viewing platform and highlight the three-dimensional landscape to enhance the value of tourists' viewing experience.
- 3) Attention should be paid to the authenticity and depth of tourists' experience of grassland and national culture. Local herders can provide services such as ethnic customs guides and yurt accommodations and develop experiential activities such as food and handicraft production, horse training, and sheep picking so that tourists can personally participate in and feel the charm of grassland culture.

1. Introduction

UNESCO recognizes World Natural Heritage site biodiversity and other natural values, as well as outstanding universal values that promote the sustainable development of human society (Chen et al., 2023). Ecosystems serve as natural substrates and physical carriers of the world's natural heritage OUV (He et al., 2024). With their complex terrains and diverse ecosystems, natural mountain heritage sites have exceptional value in biodiversity, aesthetic landscapes, cultural heritage, and educational services. They offer a unique cultural ecosystem service (CES) that provides visitors with distinctive experiences in aesthetics, recreation, education, and spiritual fulfillment. These services effectively alleviate tourists' physical and mental stress, contributing to an enhanced sense of well-being and happiness (Ko & Son, 2018; Liu et al., 2021). In 2021, the number of ecotourism tourists in China reached 2.083 billion, more than half the number of domestic tourists.

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The growing demand for and overexploitation of CES have accelerated structural changes in ecosystems, leading to the degradation or loss of essential ecosystem functions (Shi et al., 2020). The rapid development of ecotourism has driven the diversification of tourist demand and behavior (Rukeya et al., 2020), shifting the functions of mountain natural heritage sites from meeting the livelihood needs of nomadic communities to focusing on ecological regulation and the provision of CES (Schirpke et al., 2017).

However, current research on World Natural Heritage Sites focuses on the supply-side assessment of ecosystem services (Schirpke et al., 2021), exploring the capacity to regulate and support services such as habitat quality, biodiversity, water supply, etc. (Chen et al., 2024; Qiao et al., 2019). Research on CES, closely related to human well-being, remains insufficient. Moreover, the impact of tourist demand and activities on ecosystem degradation has often been overlooked, making it difficult to understand the spatial relationship between tourist activities and the environment of natural heritage sites (Pickering et al., 2020). Current heritage protection measures rely on spatial zoning restrictions, leading to issues such as uneven resource utilization, reduced ecosystem connectivity, and an increased mismatch between the supply and demand of heritage sites (Verschuuren et al., 2022). Therefore, accurately identifying the spatial distribution of tourists' CES demand is vitally important for improving the supply-demand balance at heritage sites, maximizing the benefits of spatial protection and utilization, and ensuring the sustainable development of these sites.

In recent years, people have gained a clearer understanding of the importance of CES in ensuring a high quality of life and improving their happiness index (Gao et al., 2017). Furthermore, dependence on CES has increased (Madalina & Dirk, 2023), attracting extensive research in ecology, economics, geography, and other fields. In the early stages, cultural and ecological benefits were primarily assessed through virtual markets by quantifying their material value from a supply perspective using monetization methods (Cheng et al., 2019; Guo et al., 2023). With the advancement of geographic information technology, integrating multi-source grids and social network media data has shifted CES research toward more accurate and objective non-monetary mapping studies (Czembrowski et al., 2016; Guo et al., 2022). Research has utilized data indicators such as population density, nighttime light index, and media imagery to measure the potential and actual demand for CES, including spatial patterns, trade-offs, and supply-demand matching CES (Gugulica & Burghardt, 2023; Liu et al., 2021). However, the CES demand is generated by the interaction between humans and the ecosystem (Chan et al., 2012), which includes people's perceptions and subjective interpretations of the environment (Madalina & Dirk, 2023). Consequently, scholars have employed methods such as surveys, interviews, and text mining to uncover the intangible needs of residents or visitors for aesthetics, culture, and education from a microscale perspective, focusing on aspects such as willingness to pay and participatory mapping (Cheng et al., 2019).

Quantitative evaluation still faces challenges due to the intangibility of CES and its complex non-material interest relationship with humans (Small et al., 2017). Existing data evaluation systems often overlook non-material benefits such as experiences and spiritual fulfillment. At the same time, the high cost of micro-level research and difficulties in visualization means that CES demand is still primarily viewed as a mediator for evaluating or quantifying cultural service functions, with limited exploration from the user's perspective of CES (Pan et al., 2022). However, the relationship between demand and landscape features remains under-researched. The MaxEnt model effectively addresses this gap by utilizing known point-of-interest (POI) data combined with physical and geographic variables to predict and assess the spatial distribution of study subjects (He et al., 2019).

Based on this, a comprehensive evaluation model combining GIS technology and geographic detectors is proposed, encompassing "CES demand assessment and mapping—demand and landscape relationship—heritage site conservation and development." This model

addresses the shortcomings of multi-source data in incorporating human perception factors and clarifies the spatial delineation of CES demand (Teles & Pickering, 2020). Furthermore, considering that mountain natural heritage sites exhibits high ecological vulnerability and sensitivity (Zhao et al., 2020), identifying the extent of environmental variables affecting tourist CES demand, their contributions, and interaction effects can effectively locate alternative areas with high visitor demand, thus alleviating ecological pressure.

As a typical outer suburban ecotourism area, Kalajun has experienced considerable pressure for ecological protection owing to the rapid rise in tourism (Hu & Yao, 2022). Therefore, this study selected the Kalajun area of the Tianshan World Natural Heritage Site in Xinjiang, China, as the research area and obtained first-hand data on tourist demand, POI, road networks, and so on through field investigation. Combined with social media photos and topography, landscape, ecology, and environmental data, a complete evaluation index system for tourist CES demands and influencing factors was constructed. This study aims to (1) assess tourists' perceptions of the importance of cultural services in different ecosystems, (2) scientifically quantify tourists' demand for cultural services in different ecosystems and identify their spatial distribution patterns, and (3) clarify the relationship between different CES and environmental variables and further explore the driving effect of the interaction of environmental variables on tourists' CES needs.

2. Material and methods

2.1. Study area

In 2013, the Xinjiang Tianshan Mountains were included in the World Natural Heritage List. As one of its four major heritage areas, Kalajun is in Tekes County in the middle of the Ili River Valley (Fig. 1). It has a variety of landscapes such as meadows, spruce forests, mountain grasslands, plantation surfaces, and canyons. It is the best habitat for species such as snowridge spruce and Central Asian wild fruit forests. This series of highly monopolistic and primitive resources has created a prominent natural landscape in Kalajun, making it the region with the richest biodiversity and the highest aesthetic value at the Tianshan World Natural Heritage Site (Wang et al., 2016). In addition, Kalajun, inhabited by Kazakhs, has a distinctive nomadic grassland culture and folk customs. The status of the Kalajun Tourism Area as a 5A-level scenic destination since 2016 has sparked a dramatic rise in visits, from 30,500 in 2013 to 642,900 in 2019. In 2023, as the pandemic subsided, the area attracted 530,000 visitors, yielding a tourism revenue of 74.8835 million yuan. This tourism boom has significantly enhanced job prospects for local herders, with the tertiary industry's value-added climbing by 11.4 percentage points over ten years. In contrast, the primary industry's share has dropped by 9.5 percentage points. Herders' livelihoods have diversified, shifting from nomadic to sedentary. However, rapid development, including the construction of production and living infrastructure, along with a spike in tourist numbers, has led to considerable encroachment on ecological spaces, imposing substantial stress on the ecosystem.

2.2. CES basic concept connotation and content classification

An ecosystem is a functional entity formed by the interaction between organisms and their environment. Ecosystem services refer to the benefits that humans derive directly or indirectly from ecological functions (Costanza et al., 1998), which serve to meet and sustain human survival and development needs (Daily, 2013). Among these, CES are non-material benefits co-constructed through ongoing human–ecosystem interactions. These services are closely linked to the natural substrate, cognitive systems, and cultural contexts (Cheng et al., 2019).

Natural heritage refers to geological and biological structures and landscapes of exceptional aesthetic significance, or—from scientific and

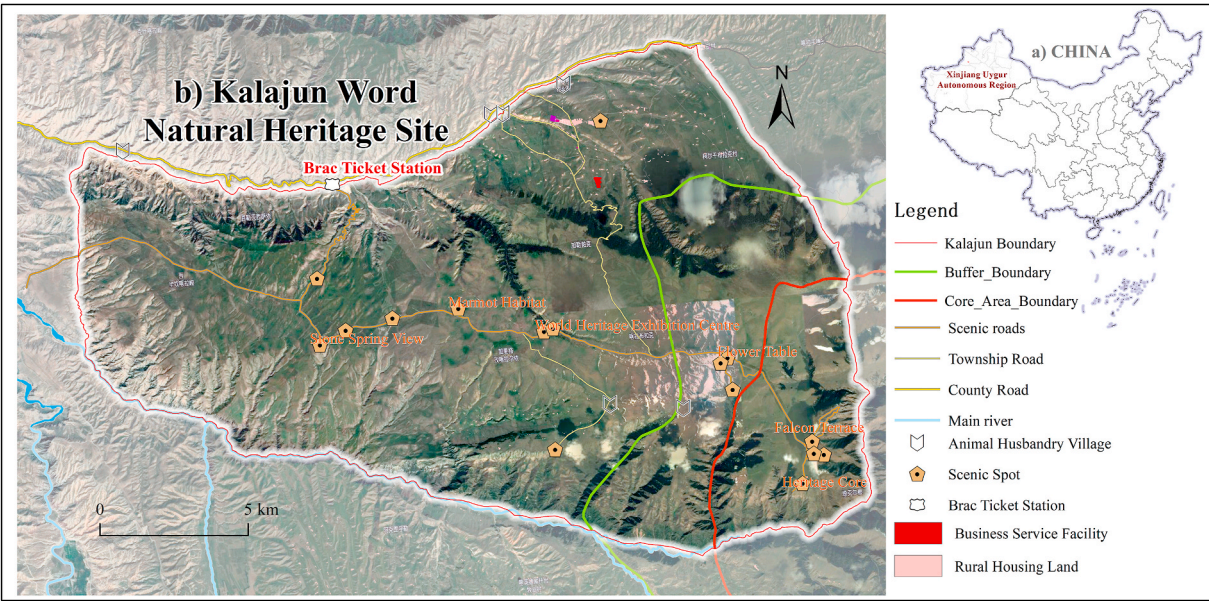


Fig. 1. Map of the study area: (a) location of the Kalajun World Natural Heritage Site, Xinjiang, China, (b) Remote sensing infographic of the Kalajun World Heritage Site.

conservation perspectives—geological, physiographic features and clearly delineated habitats of threatened species (Falk & Hagsten, 2024; Huang et al., 2025). CES reveal profound emotional, spiritual, and cultural connections between humans and nature (Luo et al., 2025). Therefore, based on the individual perceptions and preferences as well as according to the actual situation in Kalajun, this study divided the CES into five categories: aesthetics, recreation, cultural heritage, educational services, and sense of place (Table 1). The main reasons are as follows. First, the Tianshan Mountains in Xinjiang meet the World Heritage criteria (vii) and (ix), stipulating that the site must possess outstanding natural beauty and be an exceptional example of ecological and biological evolutionary processes. The Kalajun area is the most significant region, with a three-dimensional aesthetic landscape and rich

biodiversity. Second, Kalajun, once the summer pasture of the ancient Wusun Kingdom, holds considerable archaeological and historical value. It is also home to the Kazakh people, with an abundant material and intangible cultural heritage, such as nomadic pastoral traditions, falconry, and folk arts. Final, recreational and leisure services attract tourists and shape their travel experiences. A strong sense of local identity is crucial to enhancing tourist satisfaction and fostering the exchange and dissemination of local culture.

Table 1 Cultural ecosystem services and their expressions in the questionnaire.			
Types of cultural services	Definition	Expressions in this study questionnaire	Reference
Aesthetic services	The beauty that one can get in the landscape	Personally think it's a scenic place	Millennium Ecosystem Assessment (2005)
Recreation service	People choose areas to visit when they are free	An area that is relaxing and suitable for leisurely walking;	Costanza et al., 2017
Cultural heritage services	A "cultural landscape" with high historical and cultural value	Feel the atmosphere of different cultures, experience Kazakh nomadic culture, including song and dance performances, traditional costumes, and more;	Millennium Ecosystem Assessment (2005)
Educational services	A place that exhibits the diversity of flora and fauna and has educational and research value	A place where you can get knowledge of flora and fauna;	Millennium Ecosystem Assessment (2005)
Sense of place services	A special place where people can feel attached or belonging	Feel comfortable and belonging; A place that you don't want to leave, and you will miss after leaving;	Costanza et al., 2017

2.3. Methodology

First, this study collected the point data of tourists' different CES demands through participatory mapping and social media platforms and obtained the relevant environmental variable data by preprocessing the grid data. Second, the spatial distribution patterns of tourists' actual demand preferences were analyzed using the kernel density analysis tool in ArcGIS. Third, using MaxEnt software, the sample data of tourists' interest preferences and environmental variables were imported into the MaxEnt model to determine the spatial distribution patterns of tourists' different CES and to analyze the relationship between the spatial distribution of tourists' CES and environmental variables. Finally, the single key driver of the CES requirements was further analyzed using the Geodetector tool, and the interaction of the environmental variables was analyzed in detail (Fig. 2).

2.3.1. Data collection and pre-processing

- (1) As June to August is the peak tourist season in Kalajun, questionnaire surveys were conducted on-site from June 23 to 27, 2023 by a team comprising one professor and three graduate students (Table 2). Using a simple random sampling method, we targeted various categories of tourists who had completed their journeys to Kalajun, including large tour groups, small self-driving groups, and independent travelers. Before completing the questionnaire, we first asked tourists if they were familiar with the landscape patterns of Kalajun. Second, we introduced various CES's relevant concepts and connotations to tourists to effectively identify the importance and spatial distribution of tourists' perceptions of the cultural service demand for the Kalajun scenic area ecosystem. A total of 248 questionnaires were

Table 2
Scenes from the Kalajun World Natural Heritage Site and the field research.

Entrance to the scenic spot	Military Horse Culture Exhibition Hall	Observation deck
		
Vertical landscape	Questionnaire distribution site	Shepherd
		

Table 3
Principles of photo content classification.

Layers	Indicators	Landscape pattern
Aesthetic services	Natural landscape	Mountains, waters, grasslands, waterfalls, stone forests, canyons
Recreation services	Leisure activities	Hiking, tents, bicycles, horseback riding, motorcycles, off-road vehicles, telescopes, photography
Cultural heritage services	Ethnic minority culture, delicacies, folk festivals, nomadic culture, way of life, historical relics	Pastoral Road, post station, Kazakh yurt, Akon playing and singing, Kazakh felt hat, ederles, Kumiss, etc
Educational services	Wildlife	Animals (marmots, eagles), plants (grass, pine forests, flowers)
Sense of place services	Photo of nomadic production and living landscape	Pastoral areas, felt houses, grazing (e.g., sheep, cows, and horses), group photos of traditional costumes

Table 4
Raster data source and summary.

Data	Data sources	Resolution	Year
ASTER GDEM 30M	Geospatial Data Cloud (https://www.gscloud.cn/)	30 m	–
Landsat 8 remote sensing image	Geospatial Data Cloud (https://www.gscloud.cn/)	30 m	2021 年
Land use/land cover map	http://irsip.whu.edu.cn/	30 m	2020 年
NPP	(https://lpdaac.usgs.gov/)	500 m	2020 年
Meteorological data	(https://www.resdc.cn/)	1 km	2020 年
Human footprint data	(https://www.x-mol.com/)	1 km	2020 年

distributed, of which 200 were recovered and 162 were valid; the effectiveness rate of the questionnaires was 81 %. The reliability and validity of the questionnaire samples were tested using SPSS version 22.0. The Cronbach’s alpha coefficient was 0.967, indicating excellent reliability (greater than 0.9). The KMO value was 0.732, and Bartlett’s test p-value was less than 0.05, indicating good validity.

The questionnaire was divided into three parts: ① The demographic characteristics of the respondents, including sex, age, and education level; ② Tourists’ perceived importance of the five types of ecosystem cultural services. The Likert scale was divided into five levels (1 = *minimum importance*, 2 = *low importance*, 3 = *general importance*, 4 = *high importance*, and 5 = *very important*); ③ Participatory mapping demand distribution. Remote sensing image maps of Kalajun tourist attractions and roads (scenic highway and trails) were provided to tourists (photos of various small scenic spots were attached to the questionnaire for tourists to recall and correspond to the geographical locations of the scenic spots to remote sensing image maps), who marked five types of cultural service demand points on the map according to their cognition and judgment. A total of 810 valid responses were obtained.

(2) The survey focused mainly on groups of tourists with relatively limited travel times and a few self-driving tourists. To enrich the research, we will further utilize the “Two-Step Outdoor Assistant” website to obtain information on the CES needs of outdoor depth tourists who are more familiar with the research area. From November to December 2023, Using the two-step road outdoor assistant plat for(<https://www.2bulu.com/>)via a POST request method (https://www.2bulu.com/search/search_result.htm), we queried the Kalajun-related trajectory information uploaded by users. Furthermore, using selenium, the latitudes and longitudes of the summary trajectory photos were downloaded from the obtained trajectory information and imported into ArcGIS10.8, and the POI data in the non-study area were eliminated. The review eliminated ambiguities, lack of subjects, and misinformative photos, and 3886 photos were finally retained. The landscape features contained in the photos were classified to identify the CES in the photos (Brown & Reed, 2000; Oteros-Rozas et al., 2018). The evaluation classification criteria are listed in Table 3. Considering the impact of the familiarity of relevant researchers with the spatial distribution of the study area on the identification of photo content, after an in-depth investigation and understanding of the local environment, all subsequent photos were analyzed by the same researcher to consider the sensitivity of specific sites and fully capture the CES contained in the photos while not losing the internal consistency of cross-study landscape comparison. The content analysis time for each photograph was approximately 30 s.

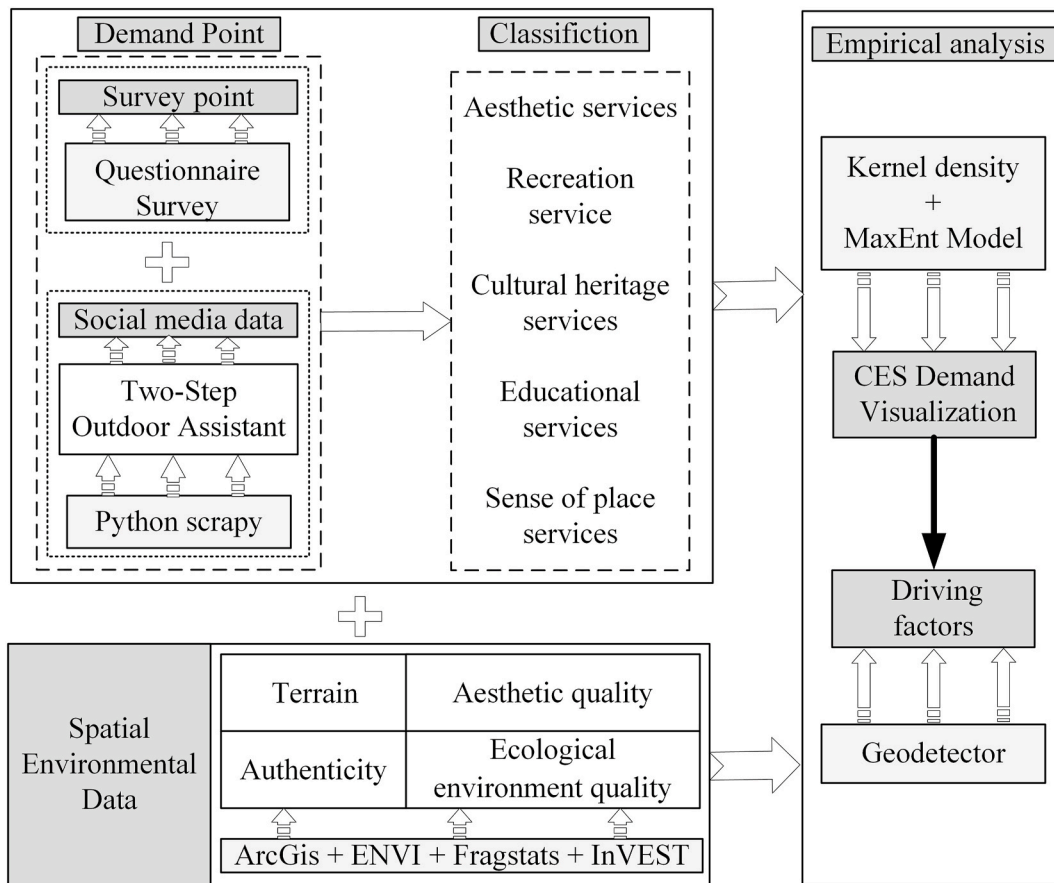


Fig. 2. The workflow diagram.

(3) Drawing on previous studies (Yang & Cao, 2022), this study improves the multi-source geospatial database, including the impact indicators of tourist demand, such as terrain, landscape, and environment, from the perspective of supply (Table 4). Owing to the limited research on the availability of grid data in the current year and the small changes in such data in a short time, the data layers in this study were mainly for 2020 and 2021 and were preprocessed using ArcGIS10.8, ENVI5.3, and Fragstats4.2.1 software to obtain the required influencing factor indicators. Finally, the scale and resolution of each indicator layer were unified (30M).

2.3.2. Visualization model

(1) Selection of environmental variables

The MaxEnt model (MaxEnt) is used to evaluate and predict the potential geographical distribution of the species (Elith et al., 2010). Compared with the traditional method, the MaxEnt model can output the spatial mapping results more objectively and effectively and quantify the influence of environmental variables on the simulation results (Loc et al., 2021).

Based on the MaxEnt model conditions and reference to CES-related research (Cheng et al., 2019), this study constructed ten environmental variables from the perspective of supply, including terrain, aesthetic quality, ecological environment characteristics, and authenticity, which affect tourist demand preferences, as candidate factors for the MaxEnt and Geodetector models (Table 5) (Shen et al., 2023; Shi et al., 2020). The terrain is a direct and significant factor affecting the spatial distribution pattern of a region. Land-use types can effectively reflect human activities and resource utilization in different areas. Altitude and slope

Table 5
Selection of environmental variables and influencing factors indicators.

Layers	Indicator	Evaluation method
Terrain	CLCD	Corine Land Cover Database (Yang & Huang, 2021)
	Elevation	Geospatial Data Cloud (https://www.gscloud.cn/)
	Slope	ArcGIS10.8 software was used to extract the slope of DEM 30 m data.
Aesthetic quality	NDVI	$NDVI = (NIR - R) / (NIR + R)$, where NIR stands for the near-infrared band, and R is the infrared band
		Fragstats4.2.1
Ecological environment quality	SHDI	Fragstats4.2.1
	Habitat quality	InVEST3.9
	Connect	Fragstats4.2.1
	Net primary production	–
Authenticity	HFP	–
	Accessibility	ArcGIS10.8 software grid cost distance method

are key variables that affect the scope of human activities and the difficulty of conducting tourism activities. Therefore, rasterized Corine land cover data (CLCD), elevation, and slope were chosen to characterize the terrain features (Zoderer et al., 2016). Second, aesthetic value and the ecological environment are important factors affecting tourists' travel experiences (Yang & Cao, 2022). The Normalized Difference Vegetation Index (NDVI) indicates surface vegetation coverage and is one of the fundamental attractors of the Kalajun grassland. The Shannon Diversity Index (SHDI) is an essential indicator of regional landscape heterogeneity. It not only reflects the uneven distribution of different landscape types, especially rare patches but is also closely related to

species diversity and represents the aesthetic quality of a landscape. Habitat quality, landscape connectivity, and net primary production (NPP) of vegetation collectively serve as proxies for assessing scenic areas' ecological health, connectivity, and vitality. Collectively, these metrics provide a comprehensive characterization of the ecological status of ecosystems (Guo et al., 2023). In addition, tourists' behavior preferences are influenced by the convenience of the destination and the density of human footprints, especially in special areas such as mountainous grasslands. Areas with poor convenience and few human visits are relatively less affected by humans, and the authenticity of natural landscapes is stronger (Zheng et al., 2022).

(2) Model building

Spatial mapping of CES demand for tourists in Kalajun was performed using MaxEnt 3.4.1. First, we sequentially imported various CES demand point data and ten closely related environmental variable data layers into the software. Subsequently, 70 % of the distribution points were randomly selected as the training dataset for model building, and the remaining 30 % were used as the testing set for model validation (He et al., 2019). Next, we assessed the model performance by constructing a comprehensive Receiver Operating Characteristic (ROC) curve to reflect its sensitivity and specificity concerning continuous variables. We used the average of ten bootstrap iterations to ensure model stability as the final model result (Phillips et al., 2017). Simultaneously, the area under the receiver operating characteristic curve (AUC) evaluates model performance, with criteria of 0.5–0.6 indicating failure, 0.6–0.7 indicating poor performance, 0.7–0.8 indicating average performance, 0.8–0.9 indicating good performance, and 0.9–1 indicating excellent performance (Fielding & Bell, 1997; Swets, 1988). The results were then imported into ArcGIS10.8 software, and the degree of demand was divided into five categories using Jenks' natural discontinuity method to obtain a demand distribution map of the CES. We analyzed the influence of different environmental variables on CES demand in the Kalajun Tourist Area using the software's built-in contribution rate and the jackknife method for verification.

2.3.3. Evaluation of driving factors

Geodetectors are tools that utilize and detect spatially heterogeneous features (Wang et al., 2010). It can effectively analyze the influence mechanism of the spatial differentiation of tourists' CES demand. To reveal the influence of a single key factor on CES demand and the common driving effect of the interaction between environmental variables on tourists' CES demand, this study examined the factor and interaction detection modules of Geodetector. The Geodetector analysis model makes no assumptions. Thus, it can effectively avoid the limitations of general statistical analysis methods when dealing with causality. This is expressed as follows (Wang & Xu, 2017):

$$q = 1 - \frac{1}{n\sigma^2} \sum_{i=1}^L n_i \sigma_i^2$$

Where q is the degree of interpretation of the spatial distribution of CES demand by the detection factor, ranging from 0 to 1. The influence of each factor on CES demand increases with the value. By comparing the q value of each factor, the interaction utility intensity between the main factors and the environmental variables can be identified. n and σ^2 denote the total number of samples and the total dispersion variance in the Kalajun region; L is the number of samples in the secondary region, and i is the i -th sample in the secondary region.

3. Results

3.1. Tourists' perceptions of the importance of CES and their demand density analysis

3.1.1. Importance perception analysis

Regarding demographic characteristics, most respondents (69 %) were from provinces outside Xinjiang. Age distribution was concentrated in the 18–30 years (43 %) and 46–60 years (20 %) age groups. Education level was primarily undergraduate (53 %) and high school degree (31 %). Monthly income was predominantly in the range of 5001–8000 RMB (53 %), and the main occupations included students (22 %) and employees in enterprises and institutions (20 %). Table 6 and Fig. 3 show aesthetic services have the highest perceived value and the smallest internal evaluation differentiation among tourists considering the CES in Kalajun (mean = 4.48; S.D. = 0.70). The importance ranking of Educational services (mean = 3.96; S.D. = 0.91) and cultural heritage services (mean = 3.96; S.D. = 0.94) were ranked the lowest in importance perceptions, which may be due to insufficient promotion of rich biological and cultural resources and relatively closed information dissemination by scenic spots. However, this may also be due to tourists paying more attention to the prominent aesthetic values of scenic spots and a lack of awareness and understanding of education and cultural heritage. It is also related to tourists' motivation to pursue green leisure travel with safe, comfortable, and entertaining experiences (Kaltenborn et al., 2020). The importance perception of the sense of place (mean = 4.12; s.d. = 0.96) and recreational services (mean = 4.11; s.d. = 0.96) was second only to that of aesthetic services. Tourists' awareness of the importance of various CES is also influenced by the environmental atmosphere of the spaces where they participate in different entertainment activities. For example, our research shows that green grasslands have made outstanding contributions in providing various CES for tourists, especially regarding the aesthetic service experience, which is highly dependent on natural landscapes such as meadows and spruce forests. Critical awareness of cultural heritage is related to cultural activities and landscapes, such as horse racing, carrying sheep, and traditional felt houses in folk activities. This significantly increases awareness of cultural heritage.

3.1.2. Analysis of demand kernel density characteristics

Fig. 4 shows that the nuclear density values were spatially characterized by 'large concentration and small dispersion' from high to low. The high nuclear density areas generally showed a 45° tilted E-shaped axis belt, gathered in the buffer zone and core area of the World Natural Heritage Site in south-central East Kalajun. This area has the most abundant stereoscopic landscapes in Kalajun and attracts the most tourists. Scattered "dot-shaped" areas along the sightseeing bus road provide various CES for tourists, especially the most widely distributed demand for aesthetic services. Moreover, these "dot-shaped" areas with lower nuclear density mainly expand circularly around military horse racing fields, world heritage exhibition centers, and marmot habitats in the central and western regions of the tourist area, providing tourists with cultural heritage service and recreational service experiences, such as horse racing culture, natural education, and species diversity. In addition, visitors can fully experience the value of nomadic culture through immersive experiences in the northern Kalajun and Kafusalang Animal Husbandry Villages.

3.2. Spatial assessment of ecosystem cultural service demand analysis

3.2.1. The spatial distribution characteristics of different types of CES requirements

Compared with the kernel density results, the MaxEnt model has the potential to predict CES demand areas, and the CES demand distribution map obtained using this model covers a larger spatial range. The results show that the geographical distribution of tourists' various CES needs

Table 6
Visitors' perceived CES importance scores and environmental contributions.

	Aesthetic services	Recreation services	Cultural heritage services	Educational services	Sense of place services
Grassland	92.40 %	77.90 %	74.80 %	85.50 %	79.60 %
Leisure Facilities	38.90 %	52.70 %	44.30 %	42.70 %	52.70 %
Folk activities	53.40 %	53.40 %	64.90 %	50.40 %	57.30 %
Traditional dwellings	54.20 %	51.10 %	58 %	51.10 %	63.40 %
World heritage exhibition center	54.20 %	48.90 %	64.90 %	66.40 %	61.10 %
Mean value	4.48	4.11	3.9	3.96	4.12
Mode	5.00	5.00	4.00	4.00	5.00
Standard deviation	0.70	0.96	0.94	0.91	0.96

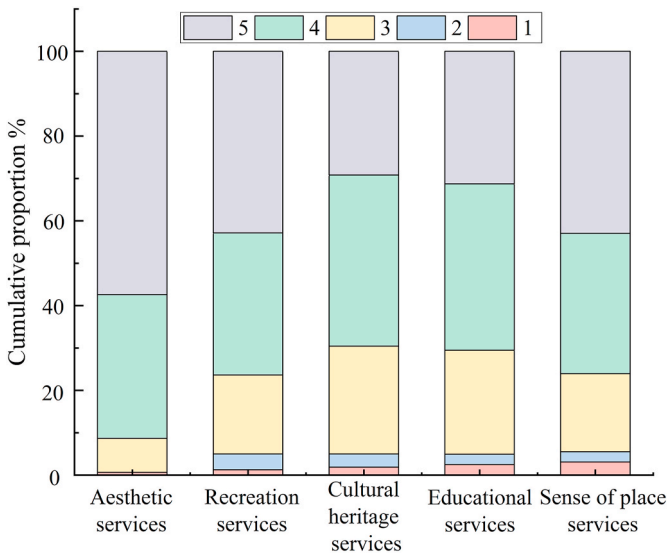


Fig. 3. Tourists' perceptions of the importance of CES.

has a certain degree of similarity, mainly because of the similar cultural, social, and economic backgrounds and needs of tourists in different regions. This makes the demand for various CES relatively concentrated in specific geographical areas, resulting in a spatial overlap between hot and cold spots.

The spatial distribution pattern of tourist demand for CES in the Kalajun Tourist Area exhibits a gradient distribution characterized by a high concentration in the central region and lower levels in the surrounding areas. Fig. 5 (a) shows that aesthetic services are mainly concentrated in East Kalajun, in a narrow “maple leaf” shape, with the highest demand area concentrated at the “petiole,” and the high demand area has directional characteristics of the sightseeing lane. Like the aesthetic services, the overlap between the distribution of recreational services (Fig. 5 (b)) and sightseeing lanes was prominent. This is due to the correlation between the two, which provide tourists with visual enjoyment and inner pleasure from relaxing experiences. More specifically, entertainment and leisure activities depend on the natural background of the comprehensive aesthetic value provided by Kalajun, which exhibits a clear homology in spatial distribution. The high-demand area for cultural heritage services (Fig. 5 (c)) is expanding to northern Kalajun Animal Husbandry Village and Kafsalang Animal Husbandry Characteristic Town based on recreational services. The areas in high demand are concentrated in the exhibition center of the Central World Heritage Site and extend eastward and westward, gradually forming a unique cultural landscape space around military and horse culture, nomadic culture, and Kazakh folk customs. Educational services (Fig. 5 (d)) showed a relatively concentrated trend in spatial distribution, with the highest-demand areas mainly located in flower beds and surrounding areas. In addition, considering individuals' emotional identification with a specific place, the high- and highest-demand areas for local tourist identity (Fig. 5 (e)) were concentrated in a strip-like pattern around the flower terraces and five flower meadows in the central part of the Kalajun region.

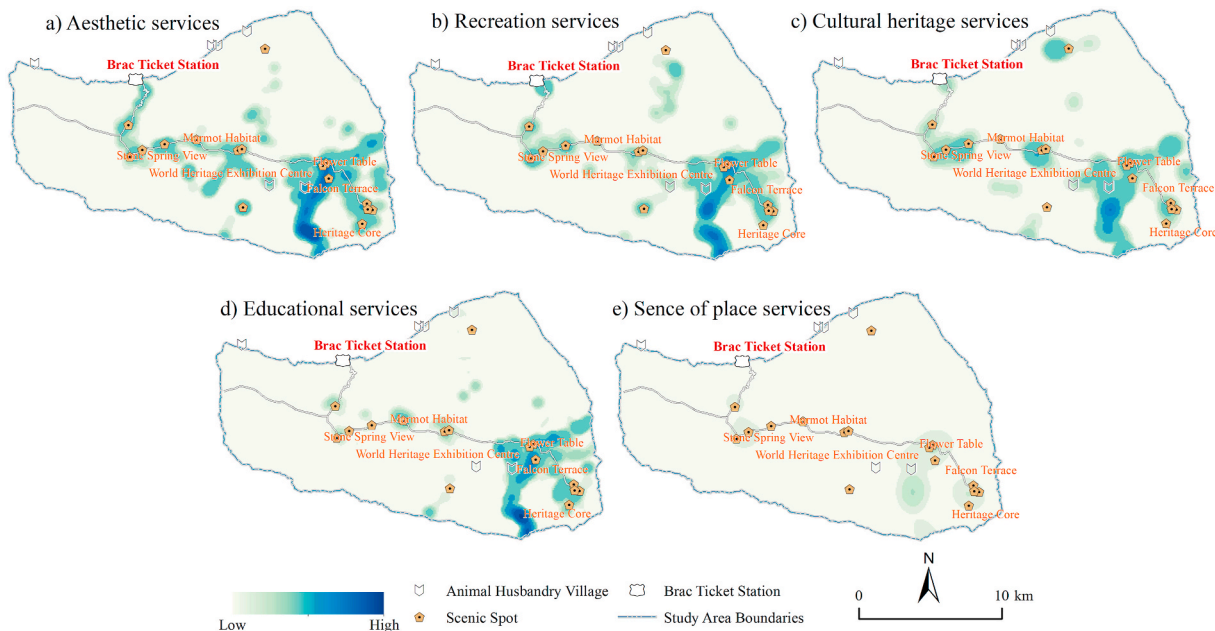


Fig. 4. Kernel density of related-CES POI.

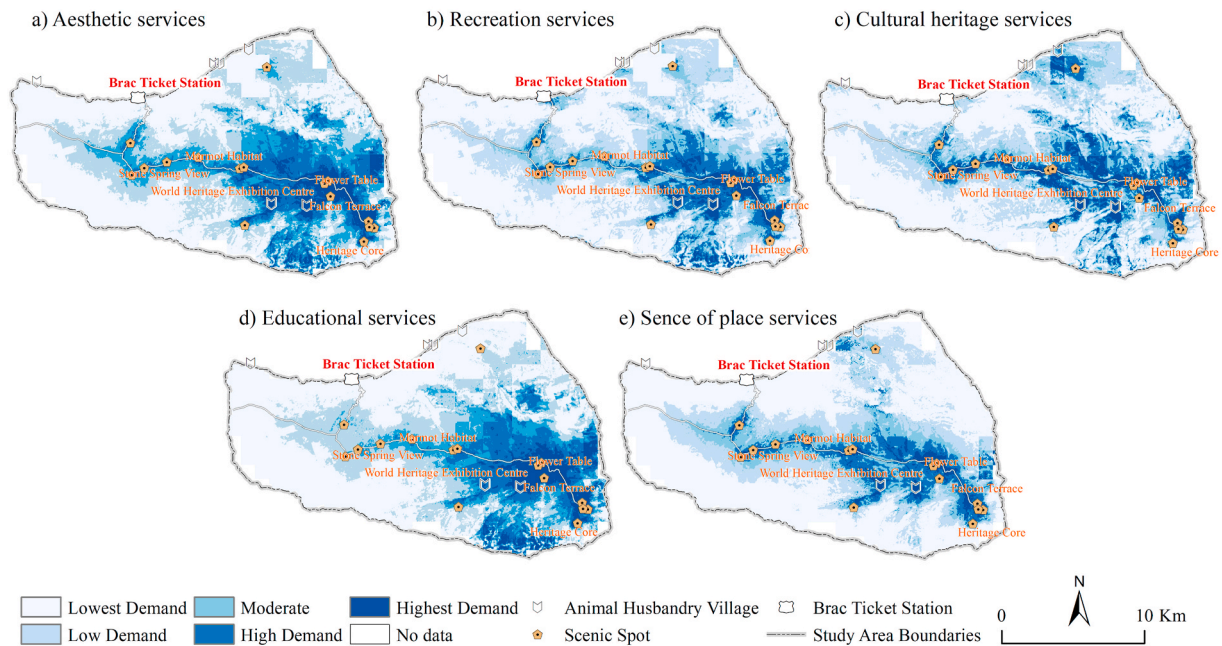


Fig. 5. Spatial distribution of five different CES demands.

3.2.2. Spatial distribution characteristics of comprehensive demand

The comprehensive demand for CES is like the spatial distribution patterns of various CES demands (see Fig. 6). However, tourists have a stronger comprehensive demand for CES regarding demand coverage, spatial range, and degree, which further indicates a significant synergistic effect of CES. Overall, the highest demand area for CES comprehensive demand was concentrated in a continuous patchy pattern in the central and southern parts of East Kalajun and expanded around the Flower Terrace and Falcon Terrace. There are also areas with a high comprehensive tourist demand and sightseeing buses. In contrast, the comprehensive demand level for CES is lower on the edge, which conforms to the law of distance decline. The minimum- and low-demand areas of CES were focused on the edge areas of West Kalajun,

accounting for 65.28 % of the total area. On the one hand, this is because Kalajun has a wide coverage area, making it difficult for tourists to reach the edge areas. However, owing to travel time constraints, most visitors join one-day or half-day tours and prefer the more diverse three-dimensional grassland landscape of East Kalajun.

3.3. Analysis of influencing factors of ecosystem cultural service demand

3.3.1. Relationship between single CES demand and environment variables

According to the prediction results simulated by the Maxent model, the AUC values of the ROC curves for various CES demands in the training and testing sets were both greater than 0.8, indicating that the Maxent model was relatively accurate in predicting the geographical

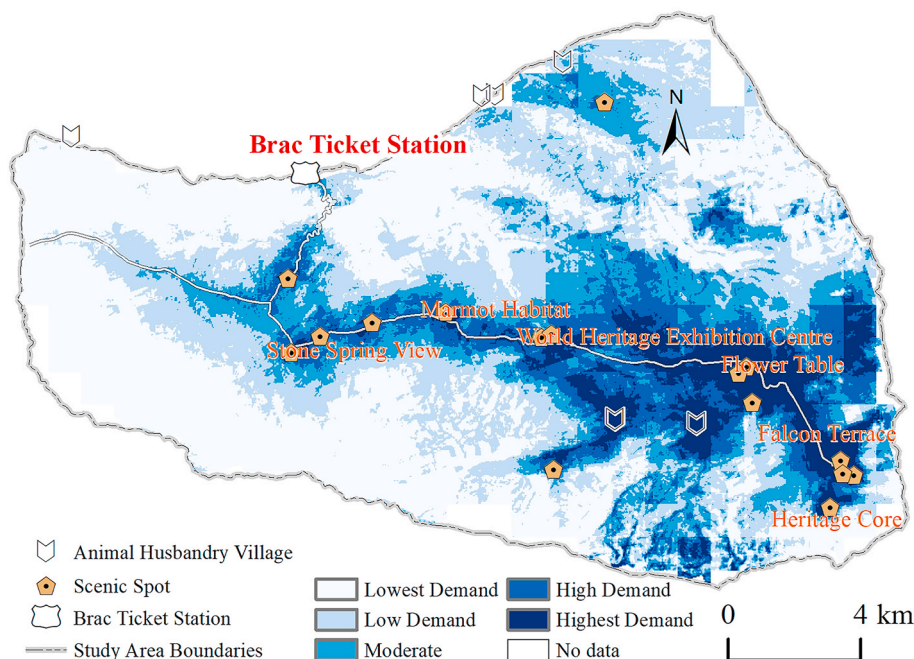


Fig. 6. Spatial distribution of visitors' comprehensive CES demands.

distribution of tourists' CES demands in Kalajun. The analysis of the relationship between CES and environmental variables in Table 7 shows that the contribution rate of the same environmental variable to different CES varies. Driven by curiosity, the public prefers landscapes with sparse human footprints and high authenticity that have not yet been developed, have a lower level of development because of the impact of transportation accessibility, or lean towards areas with convenient accessibility. The elevation and slope of the terrain are also key environmental variables that affect tourists' CES requirements. For mountain natural heritage sites such as Kalajun, although the altitude is higher, the terrain is relatively flat, and the climbing difficulty is lower. The higher the altitude, the easier it is to appreciate the full Kalajun landscape. Furthermore, Chinese tourists are eager to climb high and look far; the higher the altitude, the more it can stimulate their interest. However, for slopes, the higher the value, the greater the difficulty of climbing and the higher the physical fitness requirements for tourists, resulting in a negative impact. In addition, aesthetic value and ecological and environmental quality had a low effect on tourist demand. This is mainly because Kalajun is a core area of the world's natural heritage sites. Grassland scenery and forest canyons are extremely rich in resources, and these regions' aesthetic value and ecological environmental quality are generally high.

Additionally, based on the impact of environmental variables on the spatial distribution of tourists' CES demand, we selected the top three contributors. We created a line chart (Fig. 7) to more clearly display the relationship between each variable and the CES demand. The impact curve of altitude on the different CES needs of tourists is roughly the same, showing a trend of first increasing and then decreasing, and is mainly concentrated in the altitude range of 2400–2800 m. Beyond the sense of place, tourist demand initially diminishes and escalates as accessibility decreases. When accessibility reaches a certain reduction threshold, demand transitions from a decline to an upswing, largely because of the enhanced authenticity brought about by the scarcity of visitors. Tourists in areas with good accessibility are often more likely to gain a sense of security and belonging to a place. As accessibility decreases, the sense of unfamiliarity gradually strengthens and increases tourist tension. The human footprint was the third most influential factor, as was the direction of accessibility impact. Regions with sparse human footprints may have certain surprising landscape areas. Still, the broken line may also fluctuate because of unknown risk factors and increased difficulty in tourism, as well as tourists' individual needs and preferences.

3.3.2. Factor detection and interaction detection

Using a Geodetector to detect the driving forces of various influencing factors, the larger the q value, the more significant the driving force of the factor. The results in Table 8 show that the five factors with the most significant impact on CES demand are elevation, accessibility, slope, NPPS, and HFP. These demonstrate the close relationship between environmental variables and CES.

The interaction detection results indicated two types of interactions between the driving factors: nonlinear enhancement and dual-factor enhancement. Among these, the nonlinear enhancement accounted for 67 % of the variance. The bidirectional interaction between the various driving factors was significantly stronger than single-factor driving,

verifying the rationality of the driving factor selection. As shown in Fig. 8, the most significant interactions are Elevation $\cap$ Accessibility (0.77), Elevation $\cap$ Slope (0.60), Elevation $\cap$ Quality (0.56), Accessibility $\cap$ HFP (0.56), and Elevation $\cap$ NPPS (0.54). The results showed that the driving forces of elevation, slope, accessibility, and HFP were the largest when they interacted. Simultaneously, the effectiveness of the quality and NPPS in interacting with the ecological environment has been greatly improved. When combined with elevation or accessibility, the other driving factors were significantly enhanced, further explaining the critical impacts of terrain relief, terrain conditions, and convenience on tourists' CES demand. Combined with the divergence factor test results, the interaction of the four driving factors (elevation, slope, accessibility, and HFP) had the greatest impact on the spatial distribution of tourists' CES demand.

4. Discussion

4.1. Spatial distribution of CES demand

This study, from the perspective of individual preferences through a participatory questionnaire combined with tourist trajectory photo positioning data, can deeply understand individual perceptions and demand preferences for CES and fully determine the differences in the spatial distribution of tourists' demand for various types of CES. Different types of tourist destinations create distinct landscape experiences and atmospheres (Morse et al., 2022).

The spatial distribution of tourists' CES demand is significantly related to the spatial structure of heritage landscapes (Chakraborty et al., 2020). Like the CES demands for landscapes such as lakes and plateaus, tourists in the Kalajun tourism area prioritize aesthetic services. However, there are notable differences; visitors are drawn to the tranquility and leisure offered by wetlands and lakes, valuing the consistency of the aesthetic experiences in these environments (Yoshimura & Hiura, 2017). In contrast, for natural mountain heritage sites, tourists appreciate the diverse landscape patterns created by the interplay between grasslands, canyons, and snow-capped mountains. Natural spaces can make people feel relaxed, comfortable, and awed (Nadkarni et al., 2017; Ngulani & Shackleton, 2019) and provide a place for recreation (Peña et al., 2015). Although tourists have not given sufficient attention to cultural heritage, the grassland, nomadic, and Wusun cultures have significantly contributed to the generation of educational services and sense of place within the region's multicultural context. Scientific knowledge and nomadic cultural systems complement each other, offering visitors a richer and more profound environmental learning experience. Therefore, future efforts should focus on enhancing heritage protection and education for all stakeholders while emphasizing the authenticity and depth of visitors' experiences with grassland and ethnic cultures. Ultimately, this will enrich the cultural significance of tourist destinations.

4.2. Influencing factors of tourists' demand for CES

The CES demands perceived by tourists are influenced by various factors, including landscape type, climate, environmental variables, and personal emotions, making this relationship highly complex and nonlinear. Factors such as elevation and accessibility, which directly

Table 7
Importance analysis of influencing factors of the spatial distribution of tourist demand.

	Terrain			Aesthetic quality		Environmental quality			Authenticity	
	CLCD	Elevation	Slope	NDVI	SHDI	Quality	Connect	NPPS	HFP	Accessibility
Aesthetic services	1.36	46.20	2.78	1.39	0.22	0.86	0.06	2.64	7.75	36.74
Recreation services	1.49	28.85	5.11	2.19	0.17	1.06	0.26	5.51	10.44	44.93
Cultural heritage services	0.91	23.64	12.65	4.23	0.53	3.05	0.40	1.40	7.65	45.55
Educational services	0.92	44.63	3.95	1.54	0.74	1.06	0.20	6.93	12.04	27.99
Sense of place services	0.77	16.59	1.89	3.37	0.65	0.74	0.45	0.32	3.51	71.73

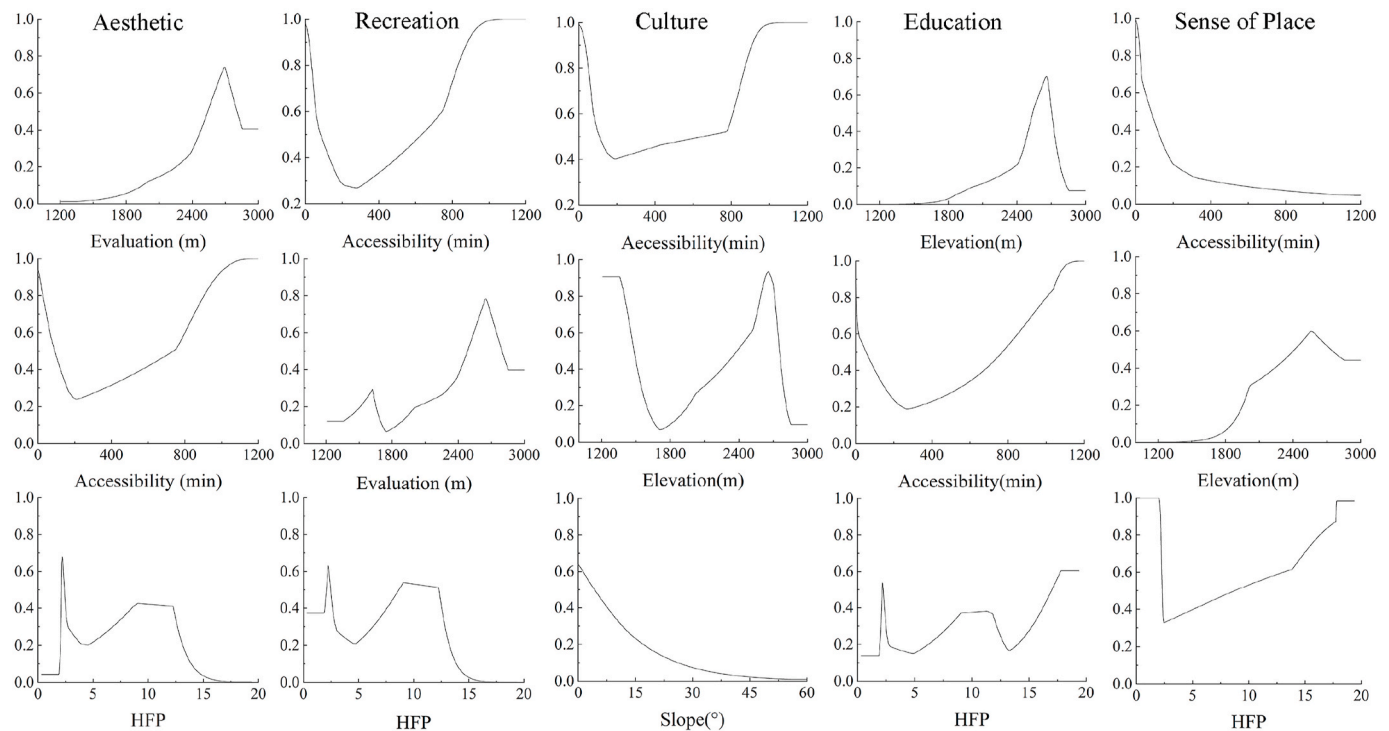


Fig. 7. Relationship between environmental variables and tourist demand.

Table 8
Geographical detection results of influencing factors.

	Elevation	Slope	NDVI	SHDI	Quality	Connect	NPPS	HFP	Accessibility
	X1	X2	X3	X4	X5	X6	X7	X8	X9
q statistic	0.440	0.145	0.119	0.010	0.024	0.008	0.140	0.140	0.344
p value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

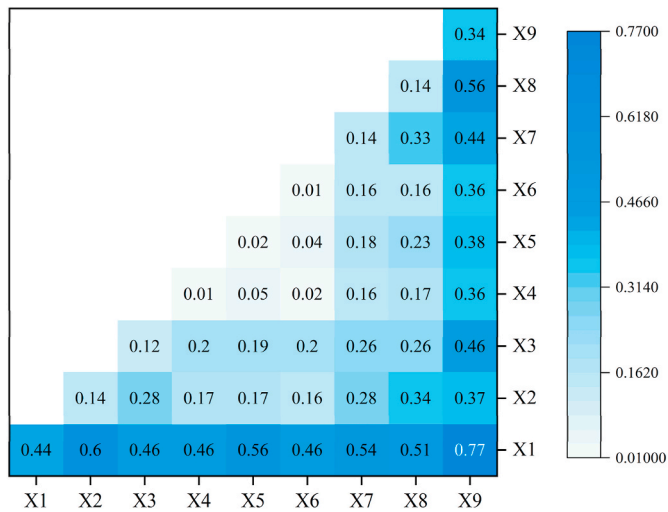


Fig. 8. Interaction of tourist CES demand driving factors.

influence visitors’ experiences and opportunities for resource acquisition, significantly impacted tourists’ CES demand at the Kalajun heritage site. However, CLCD and NDVI are key variables influencing the spatial distribution of tourists’ CES demand, which has been widely confirmed in large-scale regions, such as plateaus, forests, and lakes(Chakraborty et al., 2020; Zoderer et al., 2016). For mountain ecosystems, landscape

changes are mainly caused by “verticality” characteristic (Huertas Herrera et al., 2018). The verticality of mountain ecosystems is mainly due to their elevation(Feyissa & Gebremariam, 2018). With changes in elevation and slope, vegetation types and landscape features change, and an apparent environmental gradient forms (Huertas Herrera et al., 2018). The differences in these “vertical landscape belts” create a significant visual impact on tourists, making elevation a key variable influencing visitor demand in mountainous ecosystems. In contrast, in areas with gentler topography, such as wetlands and lakes, visual diversity primarily arises from variability in CLCD classification. Therefore, during future development and optimization, the construction of viewing platforms should prioritize enabling visitors to appreciate the unique vertical landscapes of the Kalajun ecotourism area from multiple perspectives, including forests, meadows, and artificial forests.

The Kalajun tourism area is an important ecological barrier to the Tianshan Mountains. Protecting the ecological environment of this heritage site is crucial. As a pastoral mountainous area with a sparse population and limited infrastructure, accessibility is relatively poor, with most paths being hikes or horse-riding trails. By contrast, regions near winding mountain roads offer better transportation, accommodation, and other facilities, which have a greater potential to provide related CES (Huber & Finger, 2020). This indicates an apparent directional demand for CES, with accessibility interacting significantly with the other environmental variables. Thus, in the process of tourism development and infrastructure enhancement, it is crucial to optimize the transportation network while minimizing disruption to the natural environment, thereby preserving the unique ecological characteristics and landscape of Kalajun.

4.3. Contributions and limitations of this study

The general dependence of demand on personal value perception systems poses considerable challenges to the quantitative spatial assessment of non-material interest demand (Yang & Cao, 2022). Researchers have employed socioeconomic data and value transfer models from macro and micro perspectives (Li et al., 2020; Sherrouse et al., 2014); however, the characterization of demand by socioeconomic scale data, such as the nighttime light index and population, remains controversial (Yang & Cao, 2022).

This study combines participatory mapping and social media data with location information to examine the complex behavioral preferences of tourists and explore the relationship between demand and environmental variables using multi-source spatial data. The results show that the Maxent model has high prediction accuracy, whereas the application of media data provides more data support for the spatial quantitative evaluation of CES requirements. Therefore, an evaluation framework based on a participatory mapping questionnaire, POI data, and multi-source grid data can be extended to other microscale natural heritage or ecotourism sites to provide targeted guidance and references for sustainable development.

Through the spatial integration of tourist demand, we can more accurately identify areas with tourism development potential and provide information for future management decisions regarding the Kalajun landscape (Grilli et al., 2021); however, there are still some limitations. The first is the representativeness of the social media samples. Although the number of users and visitors of the “two-step” platform has grown recently, not all visitors have used it to record their sharing tracks due to age, education, and travel motivation. Second, a spatial difference exists between the location of a photo and its representative landscape content (Yoshimura & Hiura, 2017). Climate and weather conditions are also important factors affecting relevant tourism decisions and experiences for tourists (Cao & Gao, 2022). In addition, this study found that the spatial distribution of tourist demand for the CES was closely related to the seasonal distribution pattern of the ecosystem landscape. Therefore, future research should delve into three key aspects: ① How do the differing motivations of various types of tourists—such as self-driving travelers, group tourists, and independent visitors—significantly affect their demand for CES? ② The climate of mountain natural heritage sites is variable, and how does the impact of tourism seasonality on downstream tourists’ demand for CES correlate with ecosystem landscape characteristics? ③ While considering the effect of environmental variables, we can also obtain tourists’ emotional factors through questionnaire surveys and social media photo comments to better understand people’s demand preferences and their underlying influence mechanisms.

5. Conclusion

In this study, we used participatory mapping questionnaires and tourist trajectory photo data to quantitatively evaluate the actual needs of tourists for the Kalajun CES. We explored the impact of spatial environment variables on tourists’ needs. The results show (1) tourists’ perception of the importance of Aesthetic Services > Sense of place > Recreation Services > Educational Services > Cultural Heritage Services; (2) the spatial distribution of tourists’ demand for different types of CES in Kalajun has a certain “homology” or “binding” phenomenon; (3) the comprehensive CES demand is similar to the spatial distribution of each demand, and the high demand is gradually weakening from east to west, which is consistent with the “core edge” theory and the distance attenuation theory; (4) Elevation, accessibility, and HFP play an essential role in the spatial distribution of tourist demand; (5) When combined with elevation and accessibility, the other driving factors were significantly enhanced, terrain relief, convenience, and ecological authenticity have essential impacts on tourists’ CES demand. Given this, relevant decision-makers and managers can optimize and improve the

layout of landscape facilities from the above aspects in the follow-up protection and utilization process to realize the maximization of ecological, economic, and social benefits and sustainable development of the Kalajun World Natural Heritage Site.

CRedit authorship contribution statement

Ning Shang: Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Data curation, Conceptualization. **Kaijun Cao:** Writing – review & editing, Validation, Funding acquisition, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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